Mahansh Gaur

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Computer Science (IoT) engineer and 3x national hackathon winner (incl. Smart India Hackathon 2025) specializing in Python backend engineering, AI/LLM pipelines, test automation, and embedded systems. Experienced in building production REST APIs (FastAPI, Node.js, Django), multi-agent and LLM-powered workflows with rigorous output evaluation (Gemini, OpenAI, Whisper), robust test automation (170+ tests across Vitest, Supertest, Playwright with GitHub Actions CI), and robotic computer vision (OpenCV, YOLOv8). Strong fundamentals in Data Structures & Algorithms, Object-Oriented Programming, and PostgreSQL data pipelines.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, C/C++, HTML/CSS
- **AI / ML & LLMs:** Gemini API, OpenAI API, Structured LLM Pipelines, LLM-as-Judge Evaluation, Whisper, OpenCV, YOLOv8, scikit-learn, Prompt Engineering
- **Backend & APIs:** FastAPI, Node.js / Express, Django, REST API Design, WebSockets (Socket.io), Celery, Redis, Auth (RSA-OAEP, CSRF, bcrypt, JWT)
- **Data & Databases:** PostgreSQL, MongoDB, SQLite, ETL Pipeline Design, Query Optimization, Window Functions, pandas, NumPy
- **Testing & CI/CD:** Vitest, Supertest, Playwright (E2E), React Testing Library, GitHub Actions (CI/CD Matrices), Docker, Postman
- **IoT & Hardware:** ESP32, Raspberry Pi 4B, MechArm Pi 270 (6-DOF), LoRa, MQTT, UART, Sensor Telemetry (PZEM-004T, DHT11, NPK, pH)

Professional Experience

Automatic.lly — **Jaipur, Rajasthan**
AI/ML and Data Trainee — *Jun 2025 – Present*

- **AI Backend Pipeline:** Engineered a production AI product-listing pipeline; scraped supplier URLs into MongoDB, structured outputs via Gemini 3.6 Flash Lite API, generating SEO metadata, tags, HSN codes, and descriptions for Zoho Commerce.
- **Async Architecture:** Designed FastAPI endpoints with Celery-backed async AI task execution, automated category suggestions, SKU deduplication, rate-limit resilience, and production MongoDB/Redis pooling.
- **Robotic Computer Vision:** Built autonomous pick-and-place system on MechArm Pi 270 (6-DOF arm) performing 360° scanning and HSV color segmentation to detect and sort objects in a closed loop without hardcoded coordinates.
- **Embedded Optimization:** Optimized OpenCV pipeline on Raspberry Pi 4B (2 GB RAM) at 10 FPS (320×240), reducing memory footprint by 75% using frame-skipping and capped camera buffers.
- **Visual Servoing:** Implemented proportional servoing loop (30 iterations) correcting yaw (Joint 1) and pitch (Joint 2) from pixel error; eliminated serial race conditions and hardware false readings.

AuraTech. — **Jaipur, Rajasthan**
Web Development Intern — *Jan 2025 – Apr 2025*

- Delivered Django backend routes, CRUD workflows, and real-time IoT sensor dashboards (DHT11, soil moisture, ultrasonic) integrated with live web interfaces; maintained codebase via Git pull-request review workflows.

College IoT CoE & Incubation Cell — **Jaipur, Rajasthan**
IoT Center of Excellence Intern — *Dec 2023 – Jan 2024*

- Architected end-to-end sensor data management pipeline for precision agriculture; calibrated 5 industrial IoT sensor types and engineered real-time telemetry ingestion pipelines into centralized storage.

Selected Projects

React 19, TypeScript, Node.js, PostgreSQL — **Jaipur, India**
AmitAstro: Full-Stack Consultation Platform (Solo-Built) — *Client Platform*

- Built and deployed a 30,000+ line production platform (React 19, Vite, TypeScript, Node.js, Express, PostgreSQL, Socket.io, Gemini API, PWA), directing multi-agent AI workflows and owning architecture and database schema migrations.
- Streamlined client payment verification by building UPI QR + UTR confirmation queue with instant Telegram alerts and one-tap booking unlocks.
- Enforced fraud prevention and security: 1 free call per verified phone, timezone-aware booking engine with session buffers, client-side RSA-OAEP encryption, CSRF tokens, bcrypt hashing, and rate limiting.
- Integrated real-time Socket.io chat, custom telemetry funnel (visits → signups → bookings), and Gemini-powered bilingual (English/Hindi) article generator.

Vitest, Playwright, GitHub Actions **Jaipur, India**
AmitAstro Test Automation & AI Evaluation Framework *CI/CD & Quality*

- Engineered 170+ automated tests across unit, database, REST API, WebSockets, and UI layers, executable in one command in ∼20 seconds.
- Configured GitHub Actions matrix with PostgreSQL service containers, validating cross-database behavior (SQLite & PostgreSQL) and publishing JUnit XML, coverage, and Playwright traces.
- Built security test suite guarding against SQL injection, stored XSS, forged JWTs, and IDOR vulnerabilities; fixed 6 production defects with zero flakiness across 70 repeated runs.
- Designed an AI evaluation harness with a 30-seed adversarial dataset, deterministic schema validators, and an LLM-as-judge rubric (80% human agreement) gated at 85% pass rate in CI.

Python, PostgreSQL, Docker, AWS **Jaipur, India**
ARMA: Smart Farming IoT Pipeline & Analytics Dashboard *IoT & Data Pipeline*

- Built a 5-stage Python ETL pipeline ingesting 4,300+ timestamped records from 7 IoT sensor types into a PostgreSQL star schema; cut query latency by 65% with bulk inserts and indexing.
- Derived 9 analytical features and 10 advanced SQL queries (window functions, CTEs), identifying thresholds modelling 31% water savings versus naive scheduled irrigation.
- Deployed OpenCV + YOLOv8 computer vision detecting fruit ripeness 7–10 days in advance; surfaced metrics on an AWS-hosted real-time dashboard containerized with Docker Compose.

Next.js 14, React Native, Zustand, Tailwind **Jaipur, India**
Legato: Social Music Streaming App *Full-Stack / Mobile*

- Architected a provider abstraction layer enabling instant zero-downtime switching between Audius API (free streaming, 100M+ tracks) and Spotify Web API via an environment toggle.
- Built fluid glassmorphic UI, custom audio hook for adaptive HTML5 playback, Reels-style music shorts, and social recommendations.

Honors & Hackathon Achievements

- **Winner – Smart India Hackathon (SIH) 2025** – Team Lead (Team INNOVENTORS, ID: 70380), Hardware Track (Renewable Energy). Built IoT telemetry pipeline (PZEM-004T + ESP32 + LoRa) and automated phase-balancing algorithm on distribution feeders.
- **Winner – LNMIT Ideathon 2025** – 1st Place, The LNM Institute of Information Technology, Jaipur. Presented renewable hosting capacity solution, securing second national-level win.
- **Winner – GIT Hackathon 2025** – 1st Place (AI/Data Track), Global Institute of Technology. Built AI call-quality monitoring system with Whisper speech-to-text, Gemini/OpenAI compliance scoring, n8n orchestration, and Streamlit dashboard.

Education & Certifications

Jaipur, Rajasthan
B.Tech in Computer Science Engineering (IoT) *2023 – 2027 (Expected)*
CGPA: 8.35 / 10 | **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (SQL), Machine Learning, Statistics & Probability, Computer Networks, Operating Systems.

- **Google Certification:** Crash Course on Python – Coursera (2024).
- **Languages:** English (Professional Working Proficiency), Hindi (Native / Bilingual).